

Topology of the real line – 2

Limit points, perfect sets and Cantor sets

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Theoretical minimum.

Def. A point x is a *limit point* of E if every neighbourhood of x meets $E \setminus \{x\}$, and a *condensation point* if every neighbourhood meets E in an uncountable set. The set E' of limit points is the *derived set*; E is *perfect* if it is closed and $E = E'$, that is, closed with no isolated points.

Theorem 1 (derived sets). For every $E \subseteq \mathbb{R}$ the set E' is closed, and E is closed if and only if $E' \subseteq E$. Moreover $\overline{E} = E \cup E'$, and a set F is the derived set of some set if and only if F is closed.

Theorem 2 (Cantor–Bendixson). Every closed $F \subseteq \mathbb{R}$ splits uniquely as $F = P \sqcup C$ with P perfect and C at most countable; P is exactly the set of condensation points of F . A non-empty perfect subset of $\mathbb{R}$ has cardinality continuum, so every uncountable closed set has cardinality continuum and there is no uncountable closed set of intermediate size.

Fact 1 (the Cantor set). The standard middle-thirds set C consists of the numbers $\sum_{k \geq 1} c_k 3^{-k}$ with $c_k \in \{0, 2\}$; it is compact, perfect and totally disconnected. The same construction with removed intervals of length $\alpha 3^{-k}$ produces nowhere dense closed sets of positive measure. The map $C \rightarrow \{0, 2\}^{\mathbb{N}}$ is a homeomorphism, whence C is homeomorphic to $C \times C$.

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1. Prove Theorem 1 in full, including the characterisation of derived sets, and prove the Cantor–Bendixson theorem together with the uniqueness of the splitting.
 2. (Karasev 1.54) Let $X \subseteq \mathbb{R}$ be closed and such that every $x \in X$ lies in the closure of $X \setminus \{x\}$. Prove that X has cardinality continuum. Deduce

that every uncountable closed subset of $\mathbb{R}$ contains a homeomorphic copy of the Cantor set.

3. (Karasev 1.43) Does there exist a closed uncountable subset of $\mathbb{R}$ consisting only of irrational numbers?
4. (IMC) Let $f: [0, 1] \rightarrow \mathbb{R}$ be continuous. Say that f *crosses the axis* at x if $f(x) = 0$ but every neighbourhood of x contains points y, z with $f(y) < 0$ and $f(z) > 0$.
 - a) Give an example of a continuous function which crosses the axis infinitely often.
 - b) Can a continuous function cross the axis uncountably often?
5. Prove that $C + C = [0, 2]$ and $C - C = [-1, 1]$ for the middle-thirds set C , and explain why a set of measure zero can have a difference set with non-empty interior.